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Experiment No. 05

Program :

```
#include <stdio.h>
#include <stdlib.h>
#include <conio.h>
#include <malloc.h>

typedef struct node
{
    int data;
    struct node *next;
} node;
node *createList();
node *Insert_beg(node *head, int x);
node *Delete_beg(node *head);
void PrintList(node *head);
```

```
void main()
{
    int choice, insert_option, delete_option, x;
    node *head = NULL;
    printf("Welcome to the implementation of the singly linked list ! \n");
    do
    {
        printf("Please select an operation to perform from the below list \n
        printf(" 1. Create a List \n 2. Insert a node \n 3. Delete a node \n
        printf("Enter your choice: ");
        scanf("%d", &choice);
        printf("\n \n");
        switch (choice)
```

```
        case 1:
            head = createList();
            break;
        case 2:
            do
            {
    printf(" 1 Beginning of the List \n 2. Exit the insert operation \n");
                printf("Enter your choice: ");
                scanf("%d", &insert_option);
                switch (insert_option)
                {
                case 1:
                    printf("Enter the data to be inserted: ");
```

```
                scanf("%d", &x);
                head = Insert_beg(head, x);
                break;

            case 2:
                printf("Insert operation Exit");
                break;
            default:
                printf("Please enter a valid choide: 1, 2, 3, 4");
            }
        } while (insert_option != 2);
        printf("\n \n");
        break;
```

```
    case 3:
        do
        {
printf("Select a position from where you to want to delete the element \n");
printf(" 1. Beginning of the List \n 2. Exit the delete operation \n");
printf("Enter your choice: ");
scanf("%d", &delete_option);
            switch (delete_option)
            {
            case 1:
                head = Delete_beg(head);
                break;
```

```
            case 2:
                printf("Delete Operation Exit");
                break;
            default:
                printf("Please enter a valid choide: ");
            }
        } while (delete_option != 2);
        printf("\n \n");
        break;
    case 4:
        PrintList(head);
        break;
    case 5:
```

```
        printf("Exit: Program Finished !!");
        break;
    default:
        printf("Please enter a valid choide: ");
    }
} while (choice != 5);
}

node *createList()
{
    node *head, *p;
    int i, n;
    head = NULL;
```

```
    printf("Enter the number of nodes: ");
    scanf("%d", &n);
    printf("Enter the data: ");
    for (i = 0; i <= n - 1; i++)
    {
        if (head == NULL)
        {
            p = head = (node *)malloc(sizeof(node));
        }
        else
        {
            p->next = (node *)malloc(sizeof(node));
            p = p->next;
```

```
        p->next = NULL;
        scanf("%d", &(p->data));
    }
    printf("\n \n");
    return (head);
}

node *Insert_beg(node *head, int x)
{
    node *p;
    p = (node *)malloc(sizeof(node));
    p->data = x;
    p->next = head;
```

```
    return (head);
}

node *Delete_beg(node *head)
{
    node *p, *q;
    if (head == NULL)
    {
        printf("Empty Linked List");
        return (head);
    }
    p = head;
    head = head->next;
```

```
}

void PrintList(node *head)
{
    node *p;
    printf("[ ");
    for (p = head; p != NULL; p = p->next)
    {
        printf("%d \t", p->data);
    }
    printf(" ]");
    printf("\n \n");
}
```

Output :

```
Welcome to the implementation of the singly linked list !
Please select an operation to perform from the below list
 1. Create a List
 2. Insert a node
 3. Delete a node
 4. Print the existing list
 5. Exit
Enter your choice: 1


Enter the number of nodes: 3
Enter the data: 10
20
30
```

```
Enter your choice: 1


Enter the number of nodes: 3
Enter the data: 10 20 30


Please select an operation to perform from the below list
 1. Create a List
 2. Insert a node
 3. Delete a node
 4. Print the existing list
 5. Exit
Enter your choice: 4


[ 10    20      30      ]
```

```
Enter your choice: 1
Enter the data to be inserted: 5
 1. Beginning of the List
 2. Exit the insert operation
Enter your choice: 2
Insert operation Exit

Please select an operation to perform from the below list
 1. Create a List
 2. Insert a node
 3. Delete a node
 4. Print the existing list
 5. Exit
Enter your choice: 4


[ 5     10      20      30      ]
```

```
Enter your choice: 3


Select a position from where you to want to delete the element
 1. Beginning of the List
 2. Exit the delete operation
Enter your choice: 1
Select a position from where you to want to delete the element
 1. Beginning of the List
 2. Exit the delete operation
Enter your choice: 1
Select a position from where you to want to delete the element
 1. Beginning of the List
 2. Exit the delete operation
Enter your choice: 2
Delete Operation Exit
```

```
Please select an operation to perform from the below list
 1. Create a List
 2. Insert a node
 3. Delete a node
 4. Print the existing list
 5. Exit
Enter your choice: 4


[ 20    30      ]
```